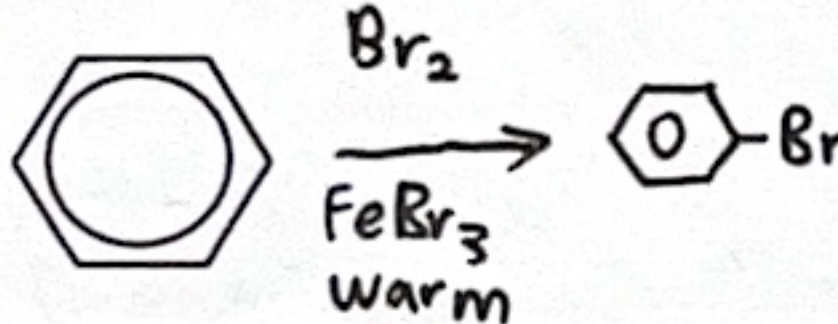
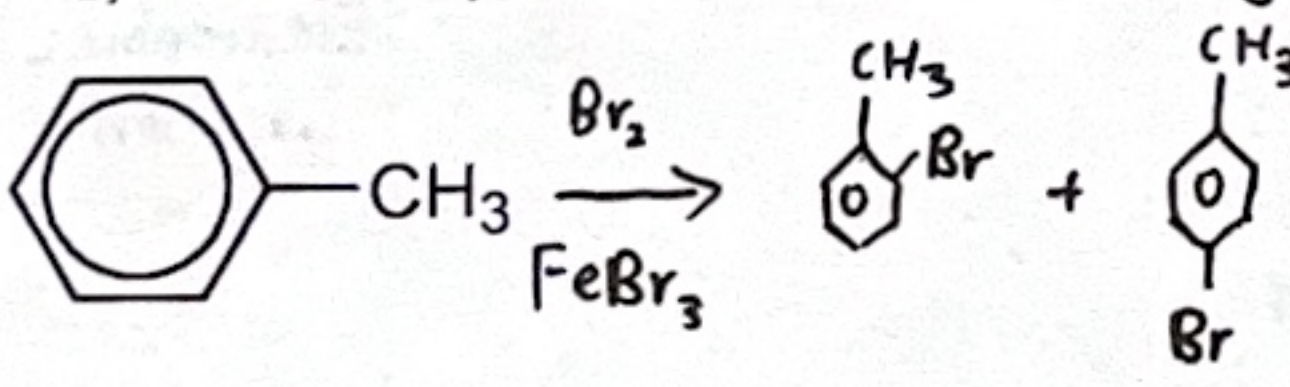
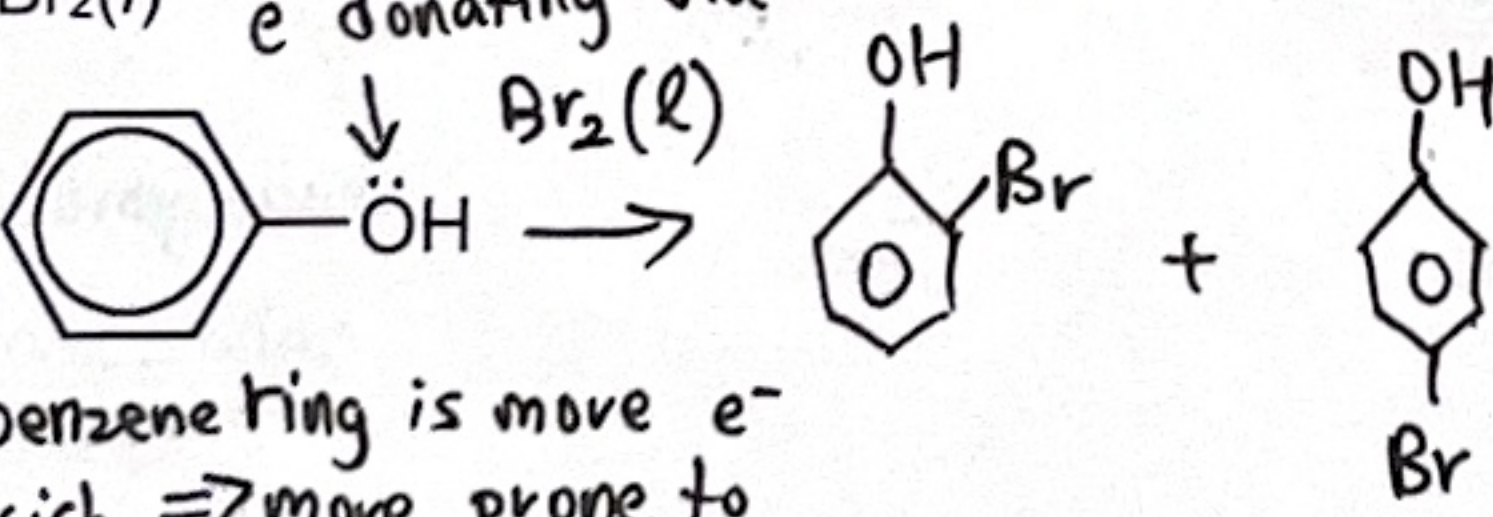
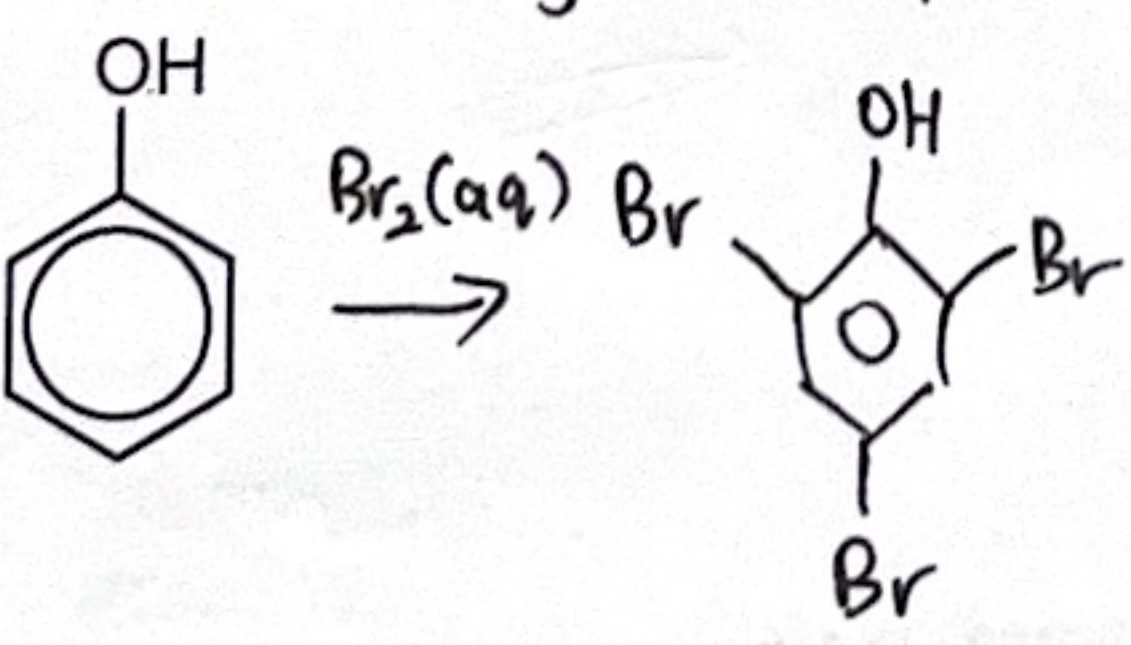
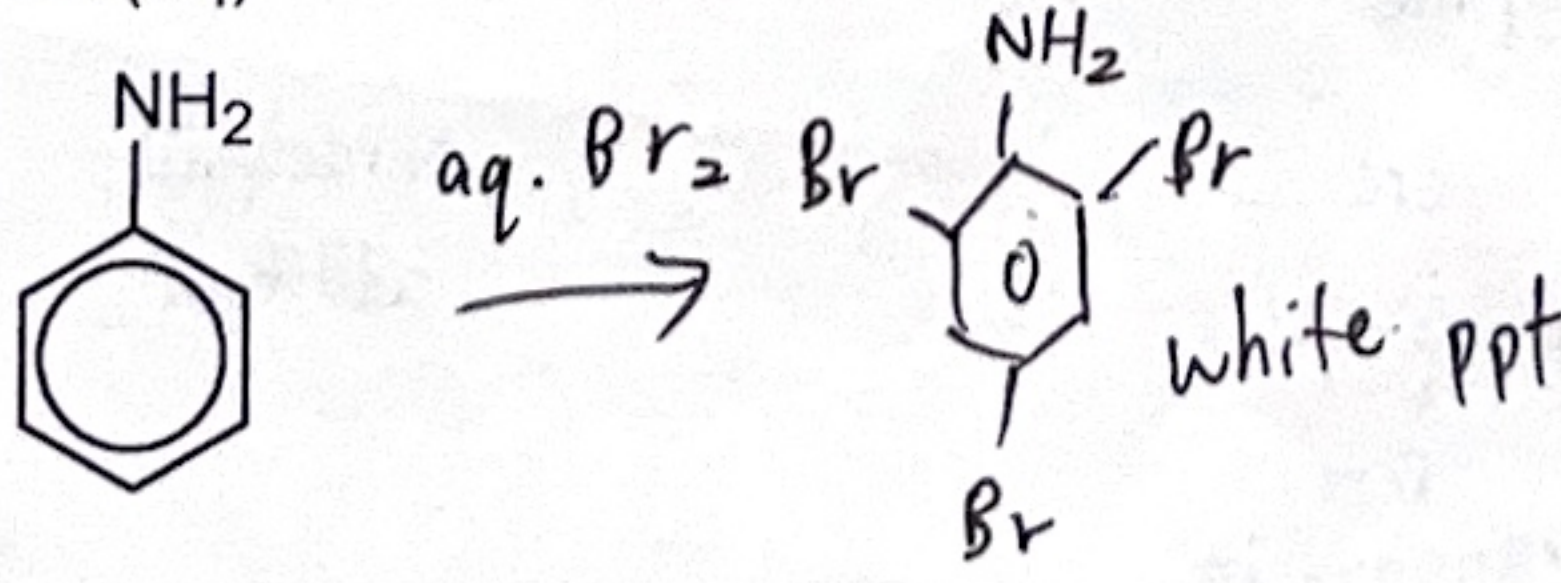


ORGANIC CHEMISTRY REACTIONS BY REAGENTS

1. Halogens, X₂

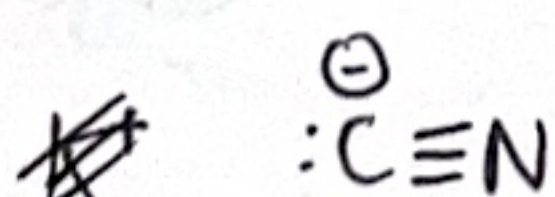
Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Alkane cannot occur with aqueous medium $\delta^+ \delta^-$ $\text{Cl}-\text{Cl}$ H_2O cannot have homolytic fission	$\text{Cl}_2(\text{g})$, uv light primary radical $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ $\text{CH}_3\text{CH}_2\text{CH}_2\text{H} \xrightarrow{\text{Cl}_2} \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \dots$ secondary radical $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_3$ for homolytic fission $\text{Cl}-\text{Cl} \rightarrow 2\text{Cl}\cdot$ Ratio is usually not 6:2 $\Rightarrow$ depends on stability of alkyl radicals	(non-polar; unreactive) Free radical substitution (alkanes are saturated)	- Leads to a mixture of products (multi-substitution). - Not suitable for synthesis.
Alkene	$\text{Br}_2(\text{l})$ $\delta^+ \delta^-$ $\text{Br}-\text{Br}$ induced dipole $\text{CH}_3\text{CH}=\text{CH}_2 \xrightarrow{\text{Br}_2} \text{CH}_3\text{CH}(\text{Br})-\text{CH}_2\text{Br}$	π bond is e^- rich Electrophilic Addition (unsaturated)	Obs: Reddish-brown Br_2 decolourised.
Alkene	$\text{Br}_2(\text{aq})$ $\text{CH}_3\text{CH}=\text{CH}_2 \xrightarrow{\text{Br}_2(\text{aq})} \text{CH}_3\text{CH}(\text{Br})-\text{CH}_2\text{Br} + \text{CH}_3\text{CH}(\text{OH})-\text{CH}_2\text{Br}$	Electrophilic addition	Obs: Yellow-orange $\text{Br}_2(\text{aq})$ decolourised.
Arene cannot be aqueous because FeBr_3 would dissociate	Br_2 , FeBr_3 , warm $\xrightarrow{\text{Lewis acid catalyst } (e^- \text{ acceptor})}$ $\xrightarrow{\text{arene is very stable}}$ 	Electrophilic substitution $\downarrow$ to maintain resonance stability	FeBr_3 is a Lewis acid catalyst which helps to e^- pair generate Br^+ acceptor - arene is very stable
Arene	Br_2 , $\text{FeBr}_3 \rightarrow$ no heat due to e^- donating alkyl group 	Electrophilic substitution	$-\text{CH}_3$ group is 2,4-directing. e^- donating/activating via inductive effect - strong e^- donating, activating
Phenol	$\text{Br}_2(\text{l})$ e^- donating via resonance effect  benzene ring is more e^- rich $\Rightarrow$ more prone to attack by electrophile	Electrophilic substitution	$-\text{OH}$ group is 2,4-directing. - Mono-substitution occurs when Br_2 is dissolved in organic solvent.
Phenol	$\text{Br}_2(\text{aq}) \rightarrow$ stronger electrophile due to water $\Rightarrow$ greater polarization of charge 	Electrophilic substitution	Obs: Yellow-orange $\text{Br}_2(\text{aq})$ decolourised. White ppt formed. insoluble
Phenylamine	$\text{Br}_2(\text{aq})$  white ppt	Electrophilic substitution	Obs: Yellow-orange $\text{Br}_2(\text{aq})$ decolourised. White ppt formed.

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Alcohol with $\begin{array}{c} \text{H} \\ \\ -\text{C}-\text{CH}_3 \\ \\ \text{OH} \end{array}$ structure $\downarrow$ [O] by iodine	$\text{I}_2(\text{aq}), \text{NaOH}(\text{aq}), \text{warm}$ $\hookrightarrow$ oxidizing agent $\begin{array}{c} \text{H} \\ \\ \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{OH} \end{array} \longrightarrow \text{CH}_3-\overset{\text{O}}{\underset{ }{\text{C}}}-\text{O}^-\text{Na}^+ + \text{CHI}_3$ yellow ppt (melts easily) $\therefore \text{temp. is } < 60^\circ\text{C}$	Oxidation	<u>Obs:</u> Yellow ppt (CHI_3) formed. - Useful as a "step-down" reaction – to shorten carbon chain by one C atom. - The carboxylic acid can be obtained by adding a strong acid e.g. $\text{H}_2\text{SO}_4(\text{aq})$ subsequently.
Aldehyde or ketone with $\begin{array}{c} -\text{C}-\text{CH}_3 \\ \\ \text{O} \end{array}$ structure	$\text{I}_2(\text{aq}), \text{NaOH}(\text{aq}), \text{warm}$ $\begin{array}{c} \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{O} \end{array} \longrightarrow \text{CH}_3-\overset{\text{O}}{\underset{ }{\text{C}}}-\text{O}^-\text{Na}^+ + \text{CHI}_3$	oxidation	

2. Hydrogen halides, HX

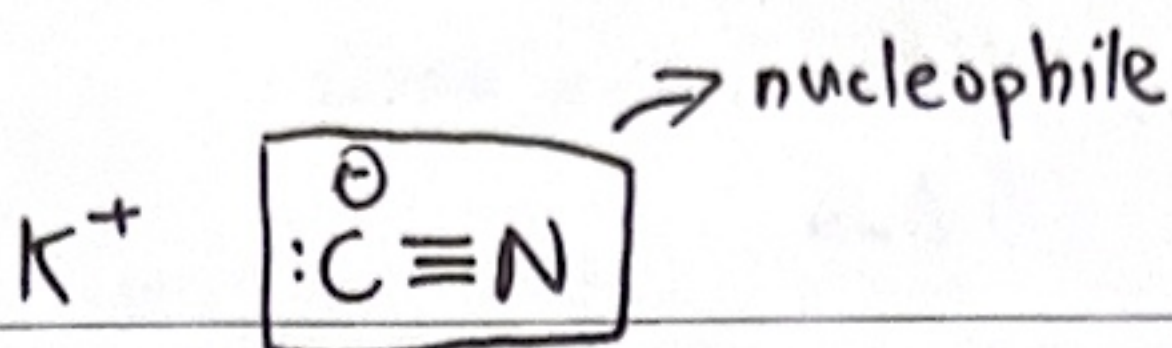
Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Alkene	$\text{HX}(\text{g}) (\text{X} = \text{Cl}, \text{Br} \text{ or } \text{I})$ $\begin{array}{c} \text{H} \quad \text{CH}_3 \\ \diagdown \quad / \\ \text{C}=\text{C} \\ / \quad \diagdown \\ \text{H} \quad \text{H} \end{array} \xrightarrow{\text{HX}} \begin{array}{c} \text{H} \quad \text{CH}_3 \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{X} \end{array}$	Electrophilic addition	Need to consider Markovnikov's rule for major product.
Alcohol	Concentrated HX ($\text{X} = \text{Cl}$ and Br), heat $\text{conc. HBr}(\text{g})$ $\text{CH}_3\text{CH}_2\text{OH} \longrightarrow \text{CH}_3\text{CH}_2\text{Br}$	nucleophilic substitution	- HBr generated <i>in situ</i> from NaBr with conc. H_2SO_4 ZnCl_2 catalyst for primary/secondary alcohols + HCl - No heating required for tertiary alcohols

3. HYDROGEN CYANIDE, HCN



Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Aldehyde or ketone	$\text{HCN}, \text{trace KCN}, \text{cold}$ $\downarrow$ weak acid source of proton $\downarrow$ catalyst, source of $:\text{CN}^-$ nucleophile $\begin{array}{c} \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{O} \end{array} \longrightarrow \begin{array}{c} \text{CN} \\ \\ \text{CH}_3-\text{C}-\text{CH}_3 \\ \\ \text{OH} \end{array}$ cyanohydrin	Nucleophilic addition	Useful as a "step-up" reaction – to lengthen carbon chain by one C atom and to increase one functional group.

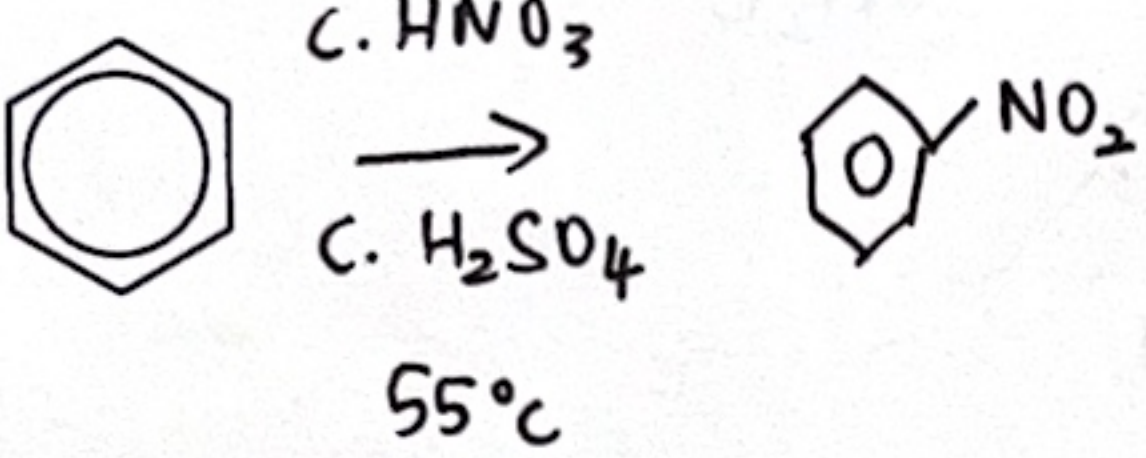
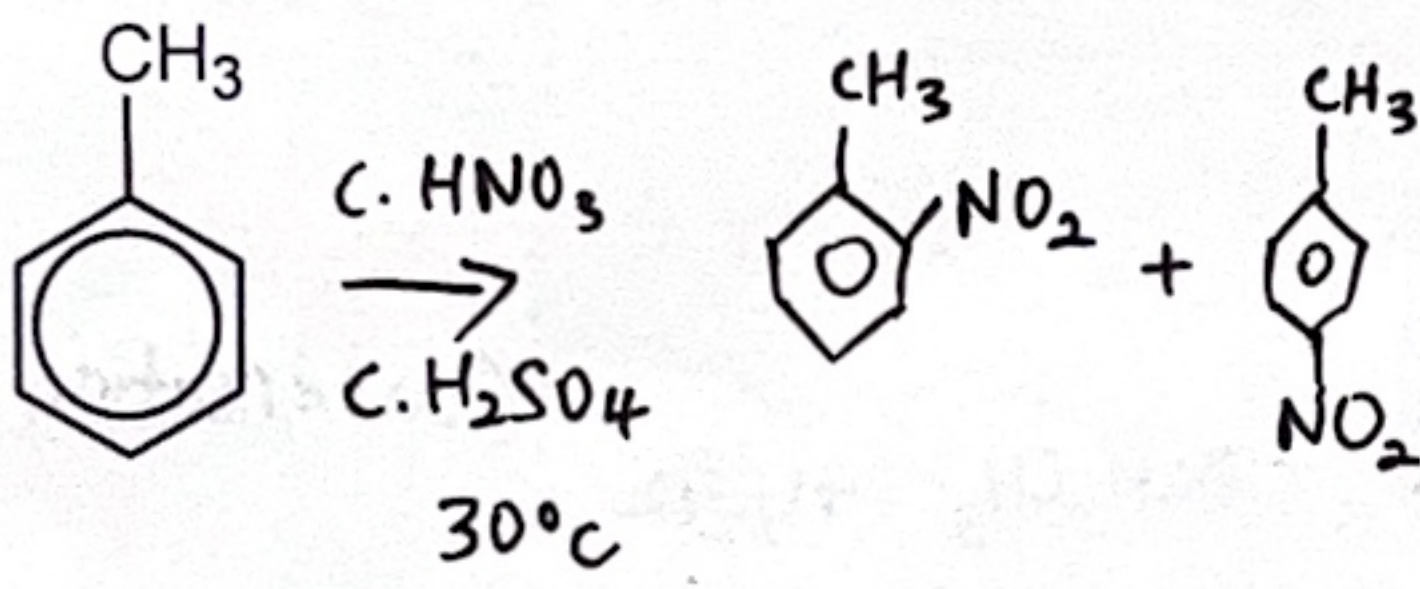
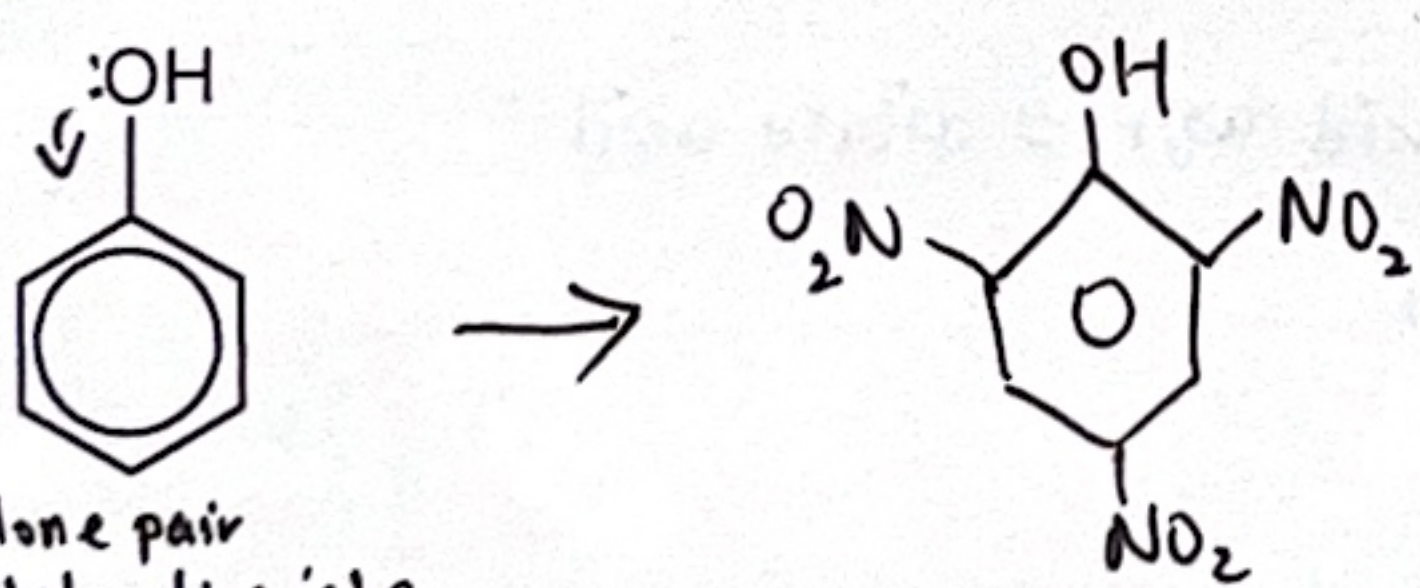
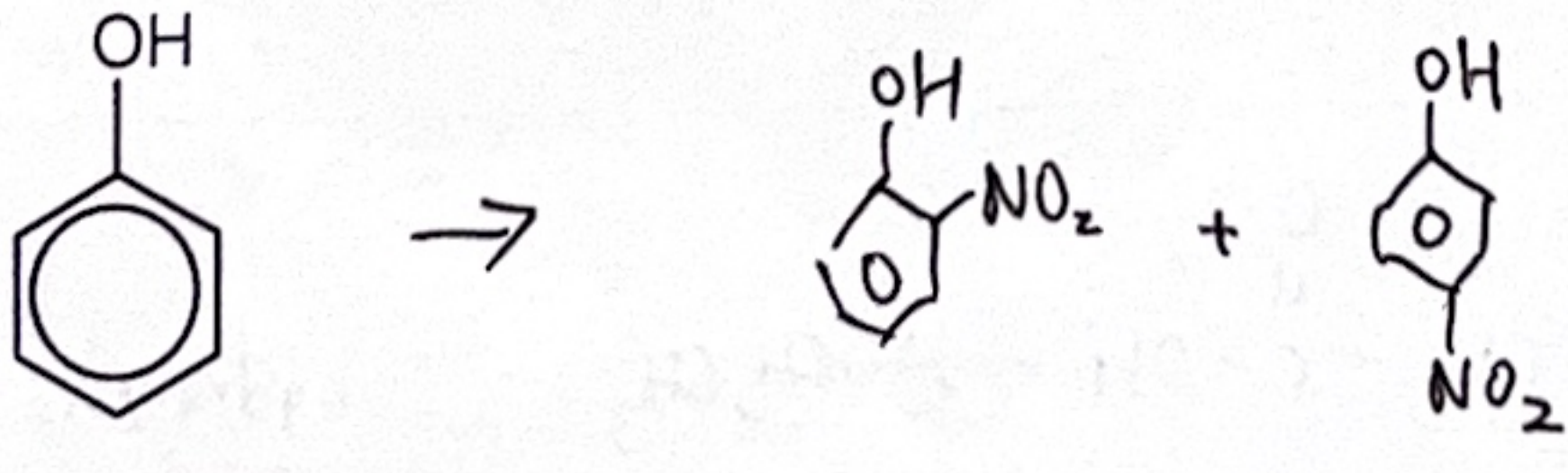
4. Potassium cyanide, KCN



Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Halogenoalkane	Ethanol KCN, heat $CH_3CH_2Br \longrightarrow CH_3CH_2C\equiv N$ nitrile	Nucleophilic substitution	Useful as a "step-up" reaction – to lengthen carbon chain by one C atom.

Note: Alcohols do not react with KCN.

5. Nitric acid, HNO₃

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Arene	Conc. HNO ₃ , conc. H ₂ SO ₄ , 55 °C 	Electrophilic substitution	Conc. H₂SO₄ is a <u>Bronsted acid catalyst</u> used to generate NO₂⁺ electrophile.
Arene	Conc. HNO ₃ , conc. H ₂ SO ₄ , 30 °C 	Electrophilic substitution	Methylbenzene requires a lower temperature for the rxn due to the presence of the electron-donating –CH₃ group. <u>Activating</u>
Phenol	Conc. HNO ₃  lone pair delocalise into ring; more e⁻ rich ⇒ more prone to electrophilic attack	electrophilic substitution	Tri-substitution occurs when conc. HNO₃ is used. ∴ higher conc.
Phenol	HNO ₃ (aq) 	electrophilic substitution	

- ① Catalyst
 ② Dehydrating agent
 ③ Acid

6. Concentrated sulfuric acid, H_2SO_4

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Alkene	1. Conc. H_2SO_4 , cold 2. water hydration $\begin{array}{c} \text{H} & & \text{CH}_3 \\ & \backslash & / \\ & \text{C} = \text{C} \\ & / & \backslash \\ \text{H} & & \text{H} \end{array} \rightarrow \begin{array}{c} \text{H} & \text{CH}_3 \\ & \\ \text{H}-\text{C}- & \text{C}-\text{H} \\ & \\ \text{H} & \text{OH} \end{array}$	Electrophilic addition industrial conditions	- Need to consider Markovnikov's rule for major product. - Alternative: steam, conc H_3PO_4 catalyst, high T, high P
Alcohol	Excess conc. H_2SO_4 , heat $\begin{array}{c} \text{CH}_3 & \text{H} \\ & \\ \text{H}-\text{C}- & \text{C}-\text{CH}_3 \\ & \\ \text{H} & \text{OH} \end{array} \rightarrow \begin{array}{c} \text{CH}_3 & & \text{H} \\ & \backslash & / \\ & \text{C} = \text{C} \\ & / & \backslash \\ \text{H} & & \text{CH}_3 \end{array} + \text{H}_2\text{O}$ $\begin{array}{c} \text{H} & & \text{H} \\ & \backslash & / \\ & \text{C} = \text{C} \\ & / & \backslash \\ \text{CH}_2\text{CH}_3 & & \text{H} \end{array}$ a few drops of	Elimination	- Conc. H_2SO_4 needs to be in excess to act as <u>dehydrating agent</u>. - Need to consider constitutional and <u>cis-trans</u> isomerism for products. - Alternative: Al_2O_3, heat
Alcohol + Carboxylic acid	Conc. H_2SO_4 , heat $\begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{C}-\text{OH} \end{array} + \text{CH}_3\text{CH}_2\text{OH} \rightleftharpoons \begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{C}-\text{OCH}_2\text{CH}_3 \end{array} + \text{H}_2\text{O}$	Condensation	Conc. H_2SO_4 is used as a <u>catalyst</u>, so only a few drops are used, making it less likely to act as a dehydrating agent. <u>for the alcohol</u>

→ dilute

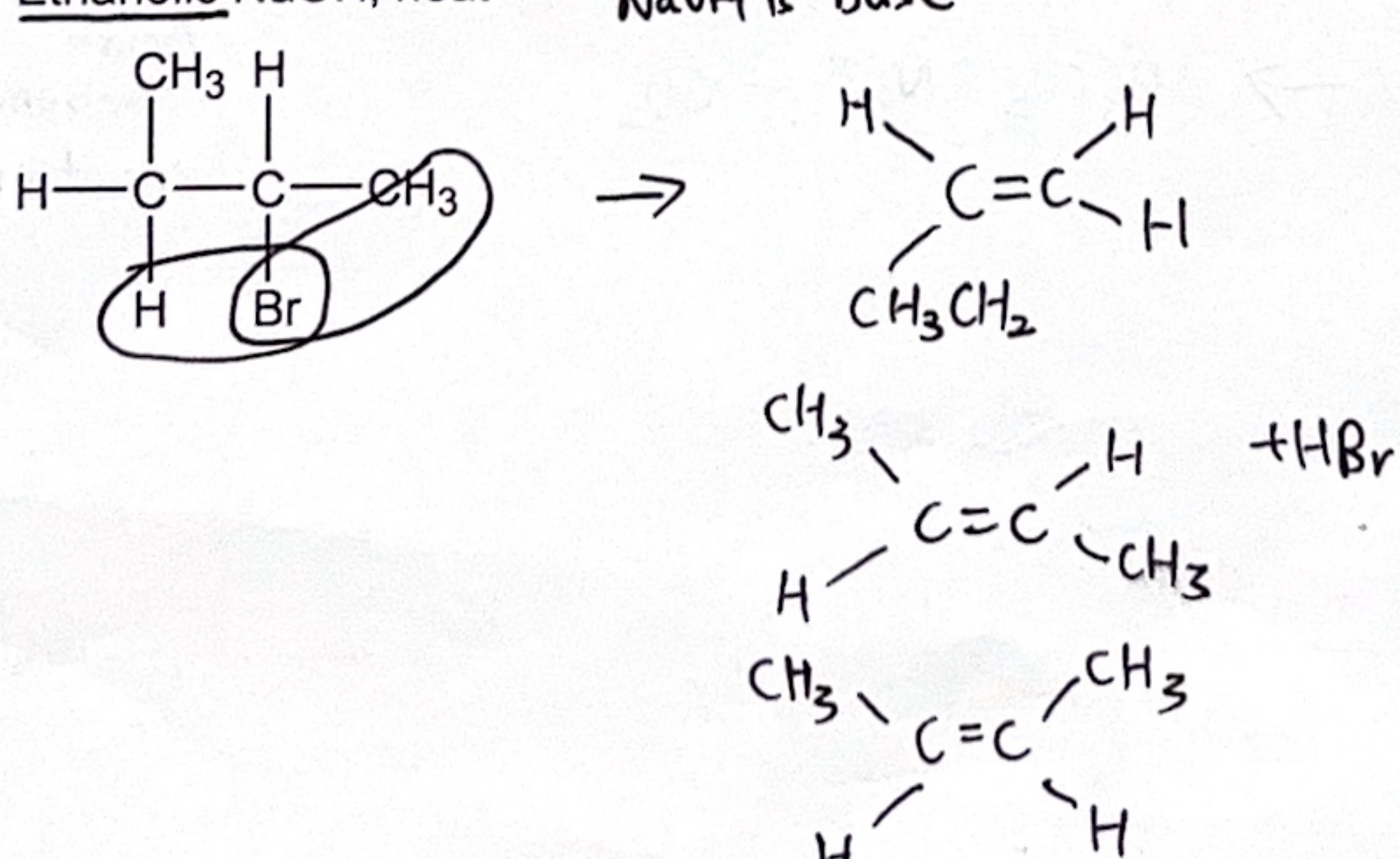
acid (aq) = dilute acid

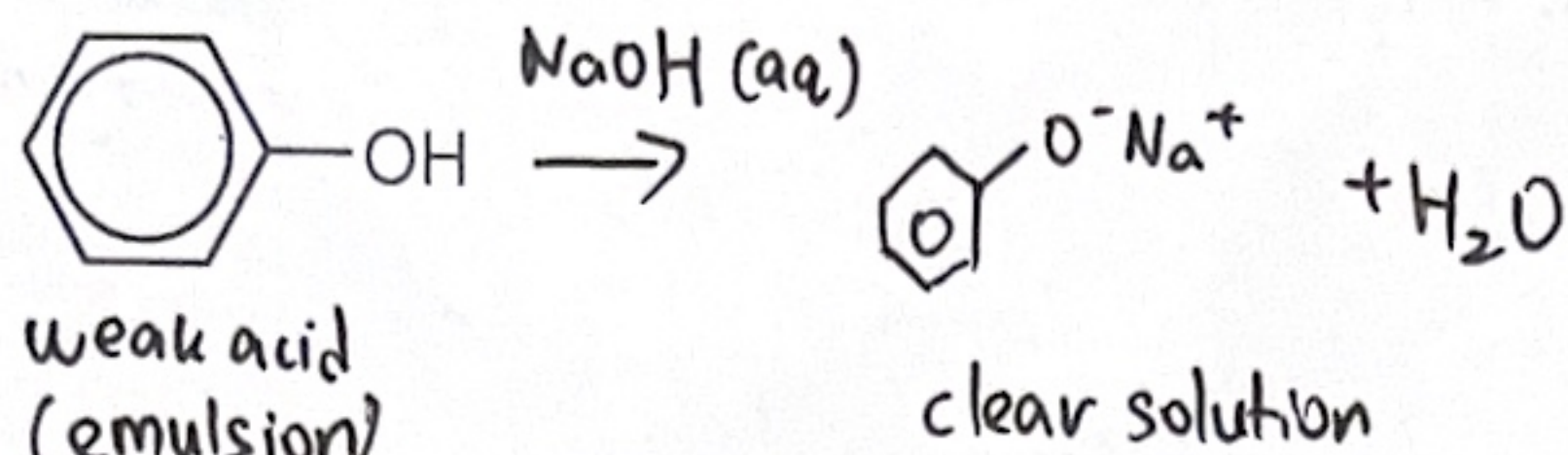
7. Dilute acids e.g. $\text{H}_2\text{SO}_4(\text{aq})$, $\text{HCl}(\text{aq})$, $\text{HNO}_3(\text{aq})$

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Nitrile	$\text{H}^+(\text{aq})$, heat $\text{CH}_3\text{CN} \xrightarrow[\text{heat}]{\text{H}_2\text{SO}_4(\text{aq})} \begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{C}-\text{OH} \end{array}$	hydrolysis	
Ester	$\text{H}^+(\text{aq})$, heat $\begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{C}-\text{OCH}_2\text{CH}_3 \end{array} \rightarrow \begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{C}-\text{OH} \end{array} + \text{HOCH}_2\text{CH}_3$ catalyst reagent is water	hydrolysis	
Amide	$\text{H}^+(\text{aq})$, heat $\begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{C}-\text{NH}_2 \end{array} \rightarrow \begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{C}-\text{OH} \end{array} + \text{NH}_4^+$ $\begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{C}-\text{NHR} \end{array} \rightarrow \begin{array}{c} \text{O} \\ \\ \text{CH}_3-\text{C}-\text{OH} \end{array} + \text{RNH}_3^+$	hydrolysis	

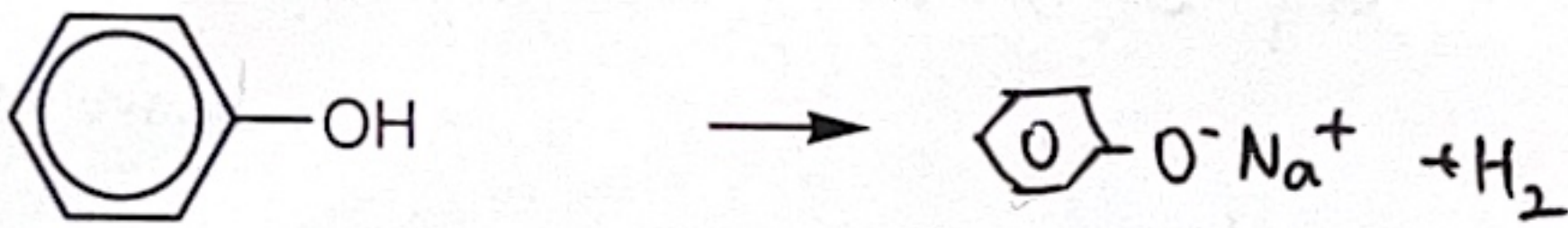
Amines basic	$\text{H}^+(\text{aq})$ $\text{CH}_3\text{CH}_2\text{NH}_2 \xrightarrow{\text{H}^+} \text{CH}_3\text{CH}_2\text{NH}_3^+$	acid base reaction	The amine can be regenerated from the corresponding salt by adding a strong base, e.g. $\text{NaOH}(\text{aq})$.
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8. Sodium hydroxide, NaOH

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Nitriles	$\text{NaOH}(\text{aq})$, heat $\text{CH}_3\text{CN} \xrightarrow[\text{heat}]{\text{NaOH}(\text{aq})} \text{CH}_3-\text{C}(=\text{O})-\text{O}^-\text{Na}^+$ carboxylic salt	hydrolysis	
Ester	$\text{NaOH}(\text{aq})$, heat catalyst $\text{CH}_3-\text{C}(=\text{O})-\text{OCH}_2\text{CH}_3 \xrightarrow[\text{heat}]{\text{NaOH}(\text{aq})} \text{CH}_3-\text{C}(=\text{O})-\text{O}^-\text{Na}^+ + \text{HOCH}_2\text{CH}_3$ reagent is H_2O	hydrolysis	
Amide	$\text{NaOH}(\text{aq})$, heat $\text{CH}_3-\text{C}(=\text{O})-\text{NH}_2 \rightarrow \text{CH}_3-\text{C}(=\text{O})-\text{O}^-\text{Na}^+ + \text{NH}_3$ $\text{CH}_3-\text{C}(=\text{O})-\text{NHR} \rightarrow \text{CH}_3-\text{C}(=\text{O})-\text{O}^-\text{Na}^+ + \text{RNH}_2$	hydrolysis	$\rightarrow$ distinguishing test, use red litmus $\downarrow$ turns blue
Halogenoalkane	$\text{NaOH}(\text{aq})$, heat OH^- is nucleophile $\text{CH}_3\text{CH}_2\text{Br} \xrightarrow[\text{heat}]{\text{NaOH}(\text{aq})} \text{CH}_3\text{CH}_2\text{OH}$	Nucleophilic substitution	If the reaction is followed by addition of excess $\text{HNO}_3(\text{aq})$ then $\text{AgNO}_3(\text{aq})$, AgX ppt will be formed. (AgCl – white ppt; AgBr – cream ppt; AgI – yellow ppt)
Halogenoalkane	<u>Ethanol</u> ic NaOH , heat NaOH is base 	Elimination	Need to consider constitutional and cis-trans isomerism for products.

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Phenol	NaOH(aq) 	Acid-base reaction	Phenol can be regenerated from the salt by adding a strong acid e.g. HCl(aq).
Carboxylic acid	NaOH(aq) $\text{CH}_3\text{CO}_2\text{H} \rightarrow \text{CH}_3\text{CO}_2^- \text{Na}^+$	Acid-base reaction	The carboxylic acid can be regenerated from the salt by adding a strong acid e.g. HCl(aq).

9. Sodium, Na

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Alcohol	Na(s) $\text{CH}_3\text{CH}_2\text{OH} \longrightarrow \text{CH}_3\text{CH}_2\text{O}^- \text{Na}^+ + \text{H}_2$	Acid-metal reaction	<u>Obs:</u> Effervescence of H_2 .
Phenol	Na(s) 	Acid-metal reaction	
Carboxylic acid	Na(s) $\text{CH}_3\text{CO}_2\text{H} \longrightarrow \text{CH}_3\text{CO}_2\text{Na}^+ + \text{H}_2$	Acid-metal reaction	

10. Sodium carbonate, Na_2CO_3 → very weak base

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Carboxylic acid	$\text{Na}_2\text{CO}_3(\text{aq})$ $\text{CH}_3\text{CO}_2\text{H} \rightarrow \text{CH}_3\text{CO}_2^- \text{Na}^+ + \text{CO}_2$	Acid-carbonate reaction	<u>Obs:</u> Effervescence of CO_2 . This reaction occurs because carboxylic acid is a stronger acid than carbonic acid, H_2CO_3.

Distinguishing
test between
alcohol and
carboxylic acid.

weak base

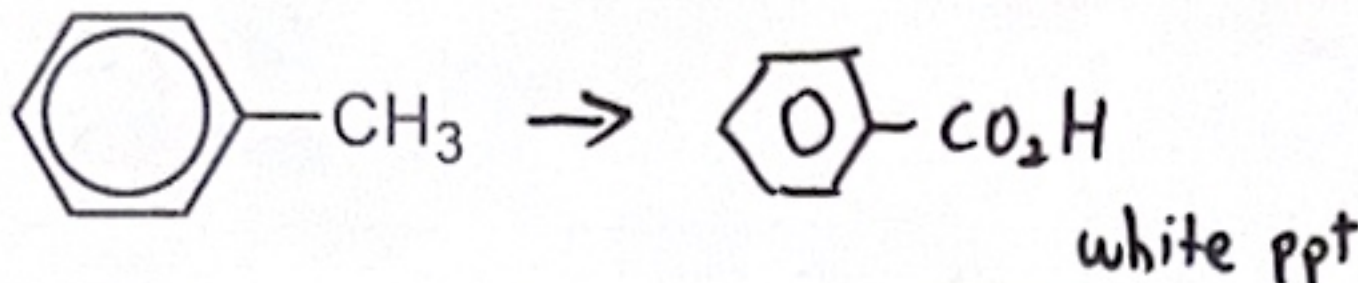
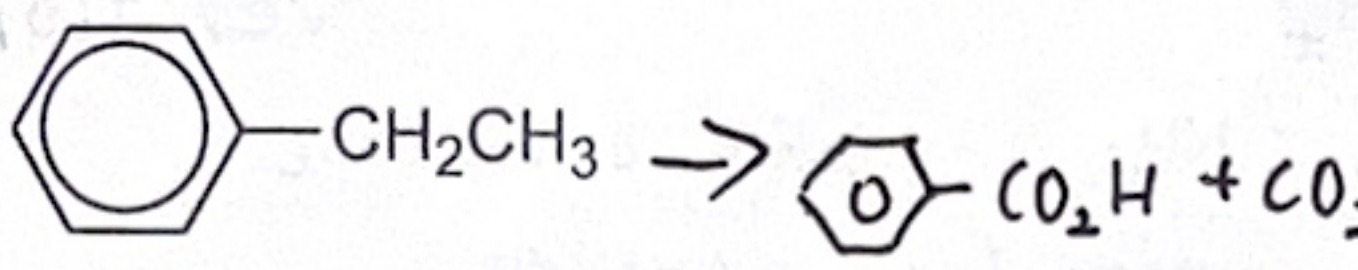
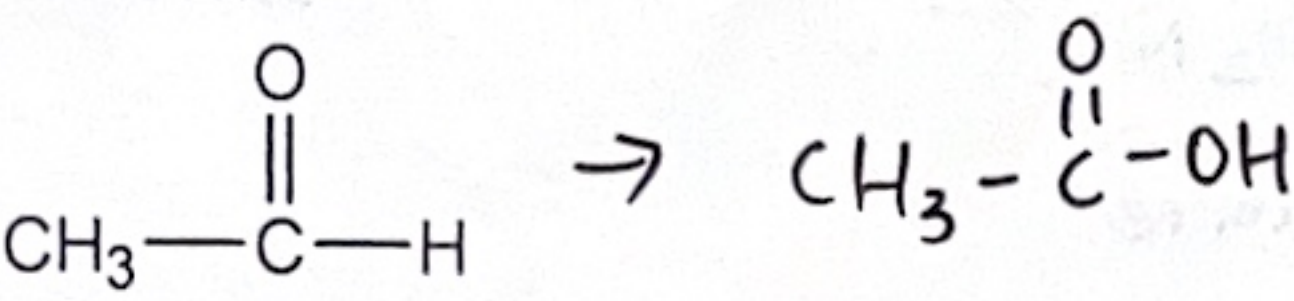
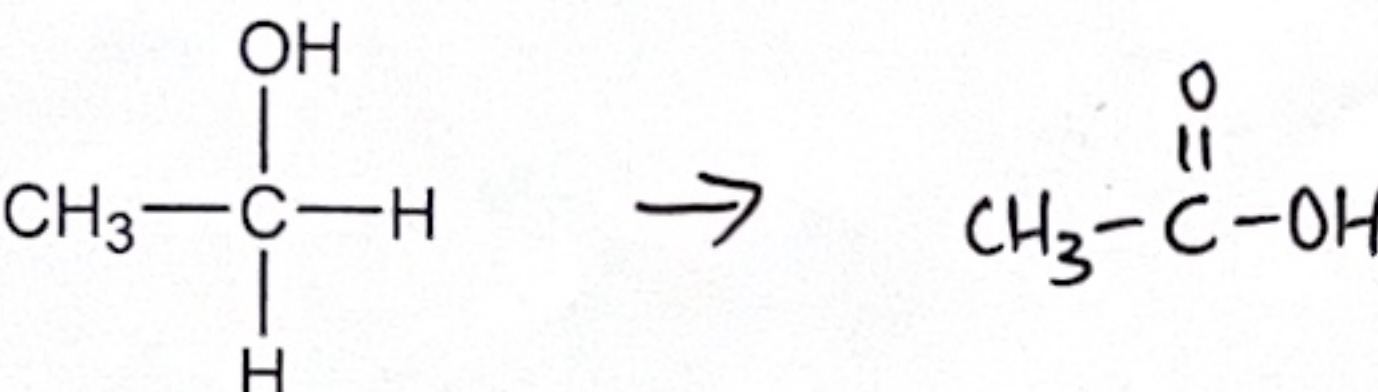
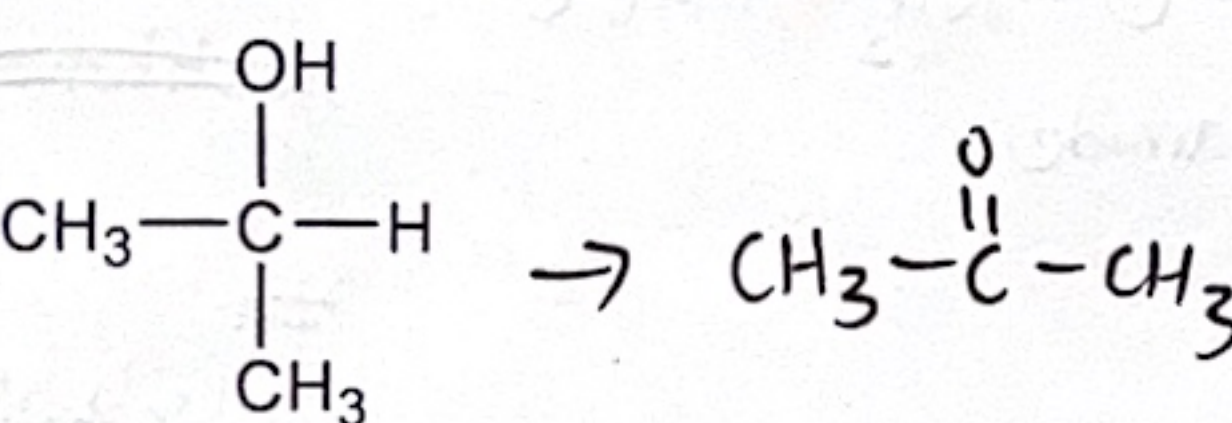
11. Ammonia, NH_3

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Carboxylic acid	$\text{NH}_3(\text{aq})$ $\text{CH}_3\text{CO}_2\text{H} \rightarrow \text{CH}_3\text{CO}_2^- \text{NH}_4^+$	Acid-base reaction	This reaction does not produce amide.
Halogenoalkane	Ethanol NH_3, heat in sealed tube under high pressure Nucleophile $\text{CH}_3\text{CH}_2\text{Br} \rightarrow \text{CH}_3\text{CH}_2\text{NH}_2$ ethanol NH_3 , heat under high pressure amine	Nucleophilic substitution	The amine produced can act as a nucleophile which can undergo further nucleophilic substitution with RX.
Acyl chloride	$\text{NH}_3(\text{l})$ or $\text{NH}_3(\text{g})$ $\text{CH}_3\text{COCl} \rightarrow \text{CH}_3\text{CONH}_2 + \text{HCl}$ amide	Nucleophilic substitution Nucleophilic acyl substitution	$\text{NH}_3(\text{aq})$ cannot be used, as the acyl chloride will be hydrolysed. - Ethanol NH_3 cannot be used as ethanol can form an ester with the acyl chloride. - Excess NH_3 used as HCl formed can react with NH_3.

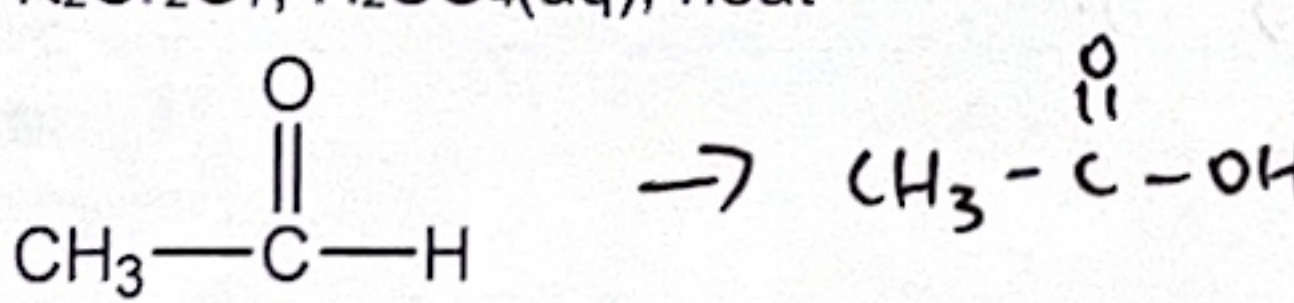
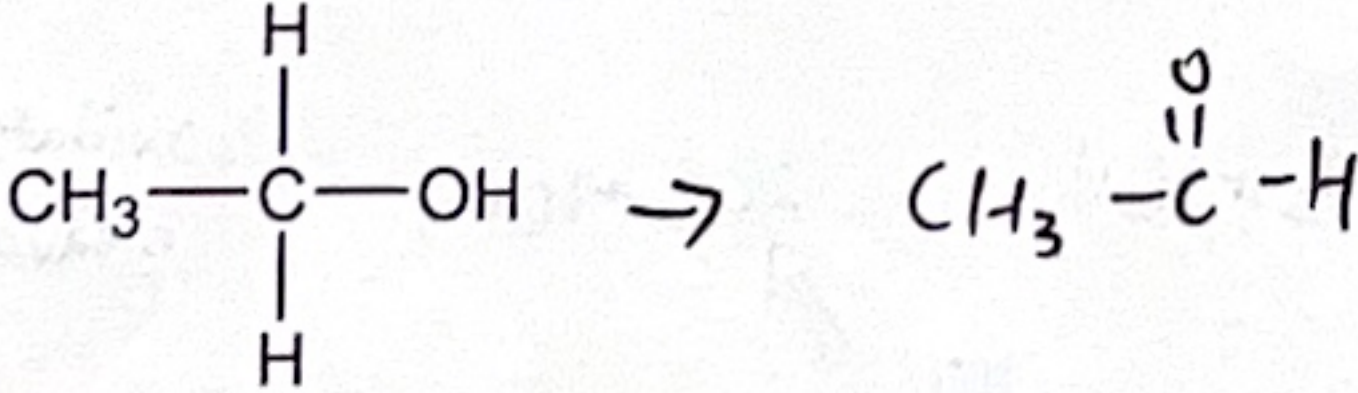
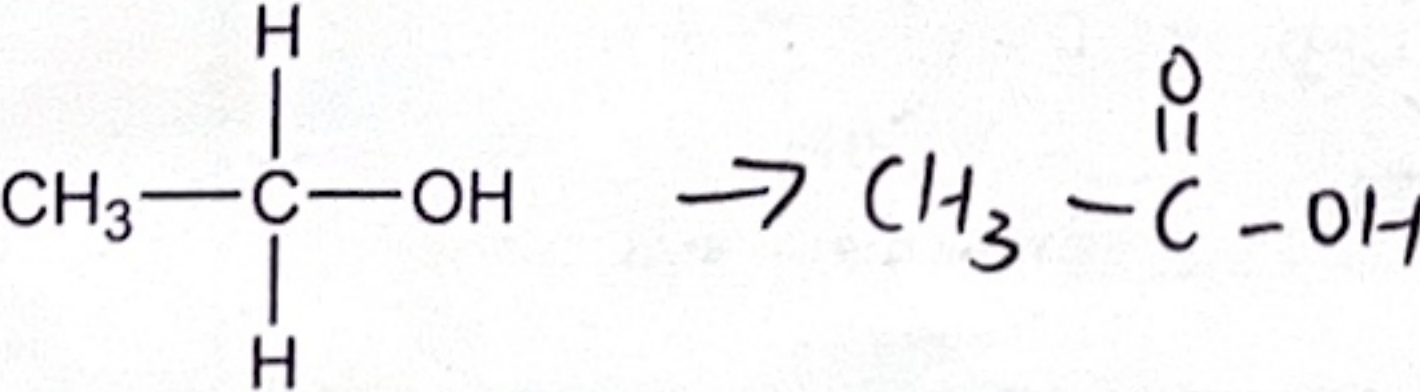
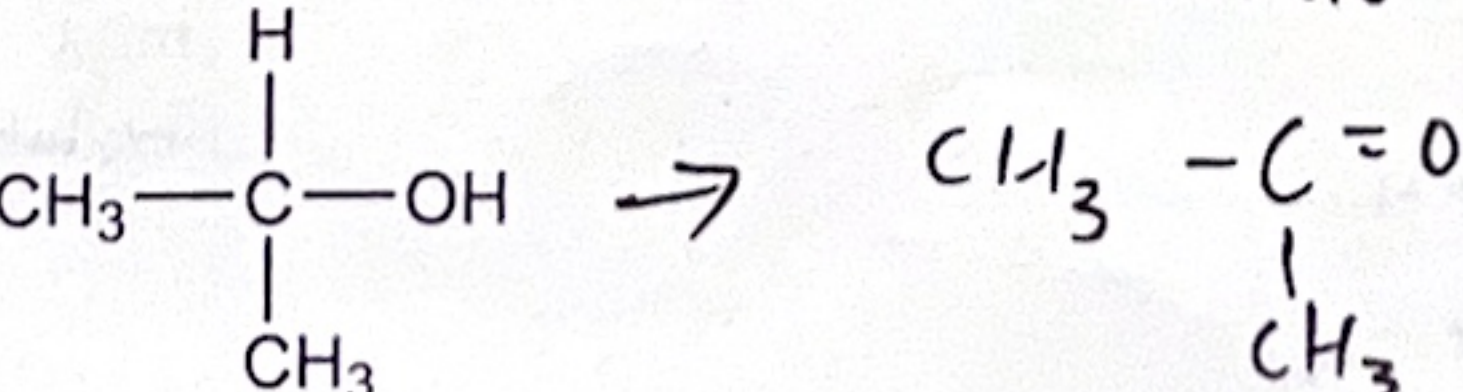
12. Potassium manganate(VII), KMnO_4 — oxidising agent

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Alkene	$\text{KMnO}_4, \text{H}_2\text{SO}_4(\text{aq}), \text{heat}$ $\text{CH}_3\text{CH}=\text{CH}_2 \rightarrow \text{CH}_3\text{C}(=\text{O})\text{CO}_2\text{H} + \text{CO}_2 + \text{H}_2\text{O}$ $\text{CH}_3\text{C}(\text{CH}_3)=\text{C}(\text{CH}_3)_2 \rightarrow 2\text{CH}_3\text{C}(=\text{O})\text{CH}_3 + \text{HOOC-COOH}$ ethanedioic acid	oxidative cleavage	<u>Obs:</u> Purple KMnO_4 decolourised. (Effervescence of CO_2 – if applicable).
Alkene	$\text{KMnO}_4, \text{NaOH}(\text{aq}), \text{cold}$ $\text{H}_2\text{C}=\text{CH}_2 \rightarrow \text{HOCH}_2\text{CH}_2\text{OH}$	mild oxidation	<u>Obs:</u> Purple KMnO_4 turned green. Brown MnO_2 ppt formed.

formed under alkaline condition

Arene	$\text{KMnO}_4, \text{H}_2\text{SO}_4(\text{aq}), \text{heat}$ 	Side-chain oxidation	Obs: Purple KMnO_4 decolourised.
Arene	$\text{KMnO}_4, \text{H}_2\text{SO}_4(\text{aq}), \text{heat}$ 	side-chain oxidation	Obs: Purple KMnO_4 decolourised. Effervescence of CO_2 .
Aldehyde	$\text{KMnO}_4, \text{H}_2\text{SO}_4(\text{aq}), \text{heat}$ 	oxidation	Obs: Purple KMnO_4 decolourised.
Primary alcohol	$\text{KMnO}_4, \text{H}_2\text{SO}_4(\text{aq}), \text{heat}$ 	oxidation	Obs: Purple KMnO_4 decolourised.
Secondary alcohol	$\text{KMnO}_4, \text{H}_2\text{SO}_4(\text{aq}), \text{heat}$ 	oxidation	Obs: Purple KMnO_4 decolourised.

13. Potassium dichromate, $\text{K}_2\text{Cr}_2\text{O}_7$

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Aldehyde	$\text{K}_2\text{Cr}_2\text{O}_7, \text{H}_2\text{SO}_4(\text{aq}), \text{heat}$ 	oxidation	Obs: Orange $\text{K}_2\text{Cr}_2\text{O}_7$ turned green. Cr^{3+}
Primary alcohol	$\text{K}_2\text{Cr}_2\text{O}_7, \text{H}_2\text{SO}_4(\text{aq}), \text{heat with distillation}$ 	oxidation	• Not used as distinguishing test.
Primary alcohol	$\text{K}_2\text{Cr}_2\text{O}_7, \text{H}_2\text{SO}_4(\text{aq}), \text{heat}$ 	oxidation	Obs: Orange $\text{K}_2\text{Cr}_2\text{O}_7$ turned green.
Secondary alcohol	$\text{K}_2\text{Cr}_2\text{O}_7, \text{H}_2\text{SO}_4(\text{aq}), \text{heat}$ 	oxidation	Obs: Orange $\text{K}_2\text{Cr}_2\text{O}_7$ turned green.

15. Hydrogen, H₂

Note: Carboxylic acids and esters cannot be reduced by H_2 .

Ether: $R-O-R$
Usually $CH_3CH_2-O-CH_2CH_3$

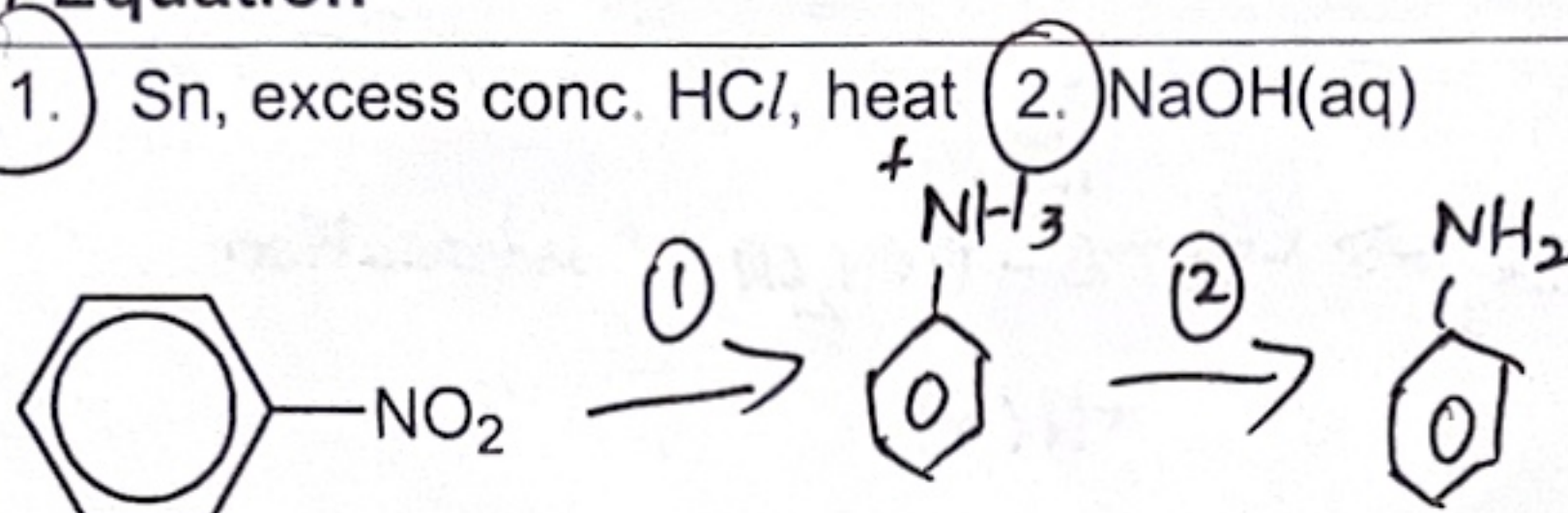
16. Lithium aluminium hydride, $LiAlH_4 \rightarrow$ highly reactive, hence no water

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Aldehyde	$LiAlH_4$ in dry ether $CH_3CHO \rightarrow CH_3-\overset{\overset{H}{ }}{\underset{\underset{H}{ }}{C}}-OH$	Reduction	
Ketone	$LiAlH_4$ in dry ether $CH_3-\overset{\overset{O}{ }}{C}-CH_3 \rightarrow CH_3-\overset{\overset{H}{ }}{\underset{\underset{OH}{ }}{C}}-CH_3$	Reduction	
Carboxylic acid	$LiAlH_4$ in dry ether $CH_3CO_2H \rightarrow CH_3-\overset{\overset{OH}{ }}{\underset{\underset{H}{ }}{C}}-H$	Reduction	
Nitrile	$LiAlH_4$ in dry ether $CH_3CN \rightarrow CH_3CH_2NH_2$	Reduction	
Amide	$LiAlH_4$ in dry ether $CH_3-\overset{\overset{O}{ }}{C}-\underset{\underset{H}{ }}{N}-H \rightarrow CH_3-CH_2-NH_2$ $CH_3-\overset{\overset{O}{ }}{C}-\underset{\underset{H}{ }}{N}-CH_3 \rightarrow CH_3CH_2NHCH_3$		
Note: Alkenes cannot be reduced by $LiAlH_4$. $\hookrightarrow C=C$ is e^- rich $\Rightarrow$ repel $:H^-$ (hydride)			

17. Sodium borohydride, $NaBH_4$ — safer

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Aldehyde	$NaBH_4$ in methanol (solvent) $CH_3CHO \rightarrow CH_3-\overset{\overset{H}{ }}{\underset{\underset{H}{ }}{C}}-OH$	Reduction	
Ketone	$NaBH_4$ in methanol $CH_3-\overset{\overset{O}{ }}{C}-CH_3 \rightarrow CH_3-\overset{\overset{OH}{ }}{\underset{\underset{H}{ }}{C}}-CH_3$	Reduction	

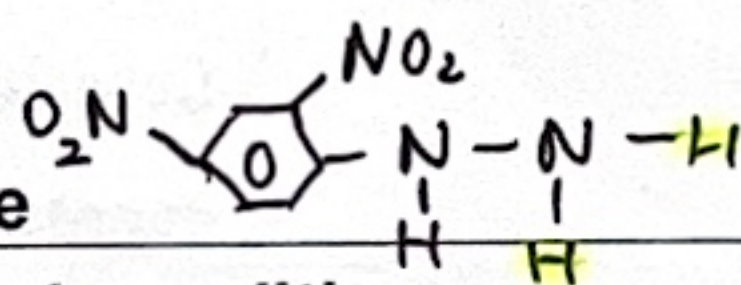
18. Tin, Sn

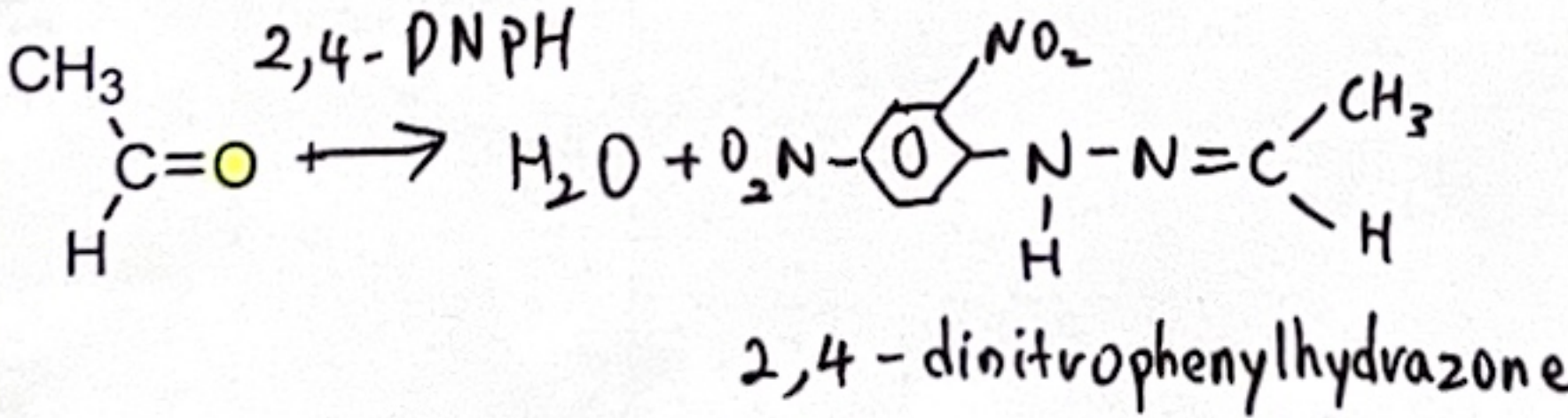
Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Nitrobenzene	1. Sn, excess conc. HCl, heat 2. NaOH(aq)  this reagent and conditions are specific to only nitrobenzene.	reduction	• Product first formed as a salt, $C_6H_5NH_3^+Cl^-$, and so NaOH(aq) is added to regenerate phenylamine.

19. Phosphorus pentachloride, PCl_5

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Alcohol	$PCl_5(s)$ $CH_3CH_2OH \rightarrow CH_3CH_2Cl$	Substitution	<u>Obs:</u> White fumes (HCl) evolved.
Carboxylic acid	$PCl_5(s)$ $CH_3CO_2H \rightarrow CH_3-C(=O)Cl$	Substitution	• PCl_3 or $SOCl_2$ may also be used. • PBr_3 or P/I_2 heat used for bromination/iodination.

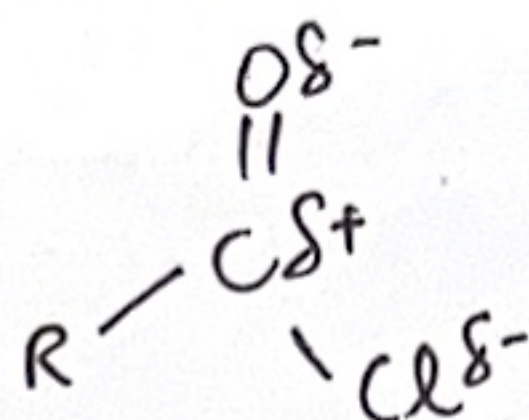
20. 2,4-dinitrophenylhydrazine



Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Aldehyde Ketone	2,4-DNPH $CH_3-C(=O)H + 2,4-DNPH \rightarrow H_2O + 2,4-dinitrophenylhydrazone$ 	Condensation	<u>Obs:</u> Orange ppt formed. Good distinguishing test as only aldehyde and ketone react

21. Water, H_2O

Functional group	Reagents and conditions / Equation	Type of reaction	Remarks
Acyl chloride	$H_2O(l)$ $CH_3-C(=O)Cl \rightarrow CH_3-C(=O)OH + HCl$	Nucleophilic sub acyl sub substitution	<u>Obs:</u> White fumes of HCl observed.

22. Acyl Chloride RCOCl 

carbonyl C is highly
e⁻ deficient $\rightarrow$ prone to
nucleophilic attack

Functional group	Eqn	Type of reaction	Remarks
Alcohol	CH_3COCl $\text{CH}_3-\overset{\text{O}}{\underset{ }{\text{C}}}-\text{Cl} + \text{HOCH}_2\text{CH}_3 \rightarrow \text{CH}_3-\overset{\text{O}}{\underset{ }{\text{C}}}-\text{OCH}_2\text{CH}_3 + \text{HCl}$	Condensation	White fumes of HCl observed.
Phenol	CH_3COCl $\text{CH}_3-\overset{\text{O}}{\underset{ }{\text{C}}}-\text{Cl} + \text{HO}-\text{C}_6\text{H}_5 \rightarrow \text{CH}_3-\overset{\text{O}}{\underset{ }{\text{C}}}-\text{O}-\text{C}_6\text{H}_5 + \text{HCl}$	Condensation	White fumes of HCl observed. • Only method to form phenyl ester • To improve yield, add Na or NaOH(aq) to phenol to generate phenoxide ion before adding acyl chloride

22. Other Reagents

$\text{Fe}^{3+} \rightarrow$ high charge density $\Rightarrow$ hydrolysis in water $\rightarrow$ acidic

Functional group	Eqn	Type of reaction	Remarks
Phenol	<u>Neutral</u> $\text{FeCl}_3(\text{aq})$ N.A.	Complex formation	Violet colouration observed. $\hookrightarrow$ complex ion